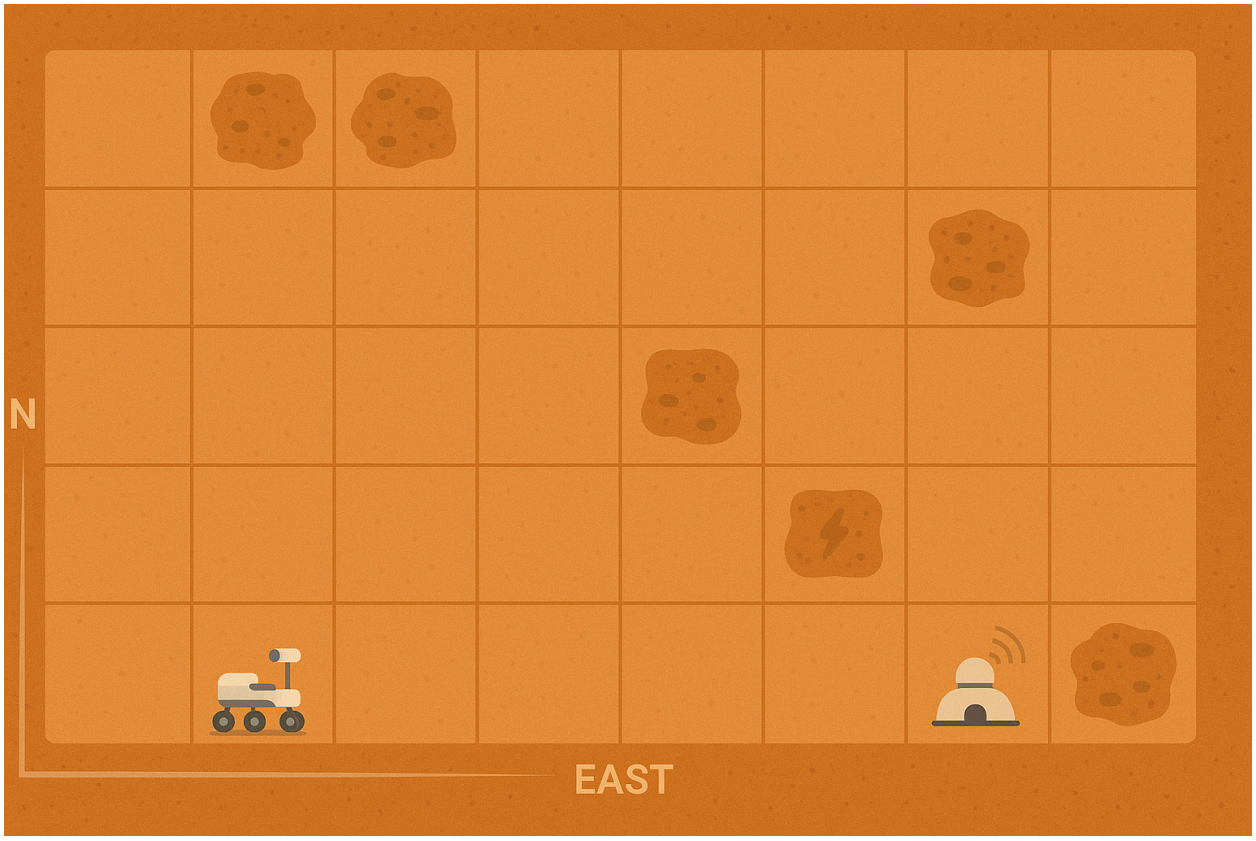


Mars Rover: Autonomous Exploration

Mission Goal:

Develop the core navigation and data collection system for "Pathfinder-Andino," our next-generation Mars Rover. Your system must enable the rover to autonomously explore a hazardous Martian region, managing its power and prioritizing scientific data collection from rare anomalies.

The Martian Plateau:

Your rover will operate on a rectangular grid-based plateau, defined by `W` (width) and `H` (height).

- Coordinates `(x, y)` start from `(0,0)` at the bottom-left. `x` increases towards the East, `y` increases towards the North.

Pathfinder-Andino Rover State:

The rover's current status is critical and includes:

- Position: `(x, y)` coordinates.
- Direction: Facing `N` (North), `S` (South), `E` (East), or `W` (West).
- Power Level: Starts at `max_power`. Power depletes with every action. If power drops to `0`, the mission fails.
- Scientific Data: Total data collected from anomalies.

Martian Terrain Features:

The `plateau_map` will contain various cell types, each with unique properties. Moving between cells consumes power, varying by terrain:

- `E` (Empty Space): Normal Terrain. Cost `1` power per move.
- `R` (Rocky Terrain): Cost `2` power per move.
- `X` (Impassable Obstacle): Rover cannot enter these cells.

- `C` (Solar Charging Station): If the rover is on this cell, it gains `charging_rate` power units per "turn."
- `A` (Scientific Anomaly): If the rover moves onto this cell, it can "collect" the anomaly, gaining scientific data.

Rover Command Protocol:

The rover receives a sequence of single-character commands as a string. Each command is processed one after the other.

`L`: Rotate 90 degrees Left (e.g., N -> W). Consumes `1` power.

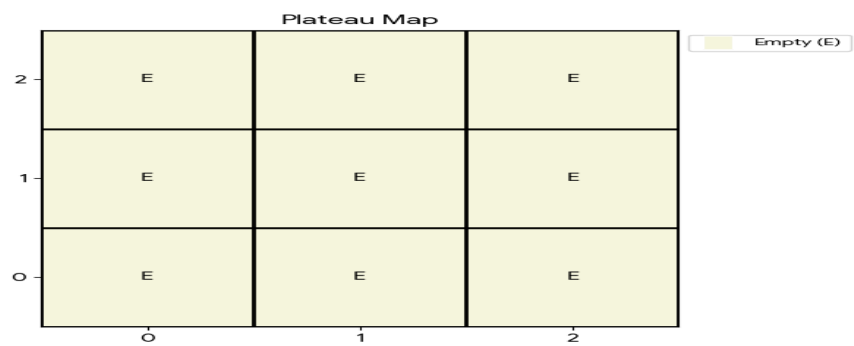
`R`: Rotate 90 degrees Right (e.g., N -> E). Consumes `1` power.

`M`: Move forward one grid point in the current direction. Consumes/Charges power based on terrain.

`P`: Process (collect) anomaly. Valid only if the rover is on an 'A' cell. Consumes `1` power per turn.

📚 List of Possible Sample Inputs and Outputs

- Test Case 1: Simple movement, no obstacles




```
{
 "plateau_map": [
  ["E", "E", "E"],
  ["E", "E", "E"],
  ["E", "E", "E"]
 ],
 "initial_state": {
  "position": [0, 0],
  "direction": "N"
 },
 "max_power": 5,
 "charging_rate": 1,
 "command_sequence": "MMRMM"
```

}

Output:


```
{
  "final_state": {
    "position": [2, 2],
    "direction": "E",
    "power": 0,
    "scientific_data": 0
  },
  "status": "Mission Successful"
}
```


- **Test Case 2: Running out of power mid-processing**


```
{
  "plateau_map": [
    ["E", "A"],
    ["C", "E"]
  ],
  "initial_state": {
    "position": [0, 0],
    "direction": "N"
  },
  "max_power": 3,
  "charging_rate": 1,
  "command_sequence": "MPPP"
}
```


Output:

{
  "final_state": {
    "position": [0, 1],
    "direction": "N",
    "power": 0,
    "scientific_data": 0
  },
  "status": "Mission Failed: Power Depleted During Anomaly Processing"
}

.....

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